

Model Paper - Database Management Systems

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Q1 (20 marks)

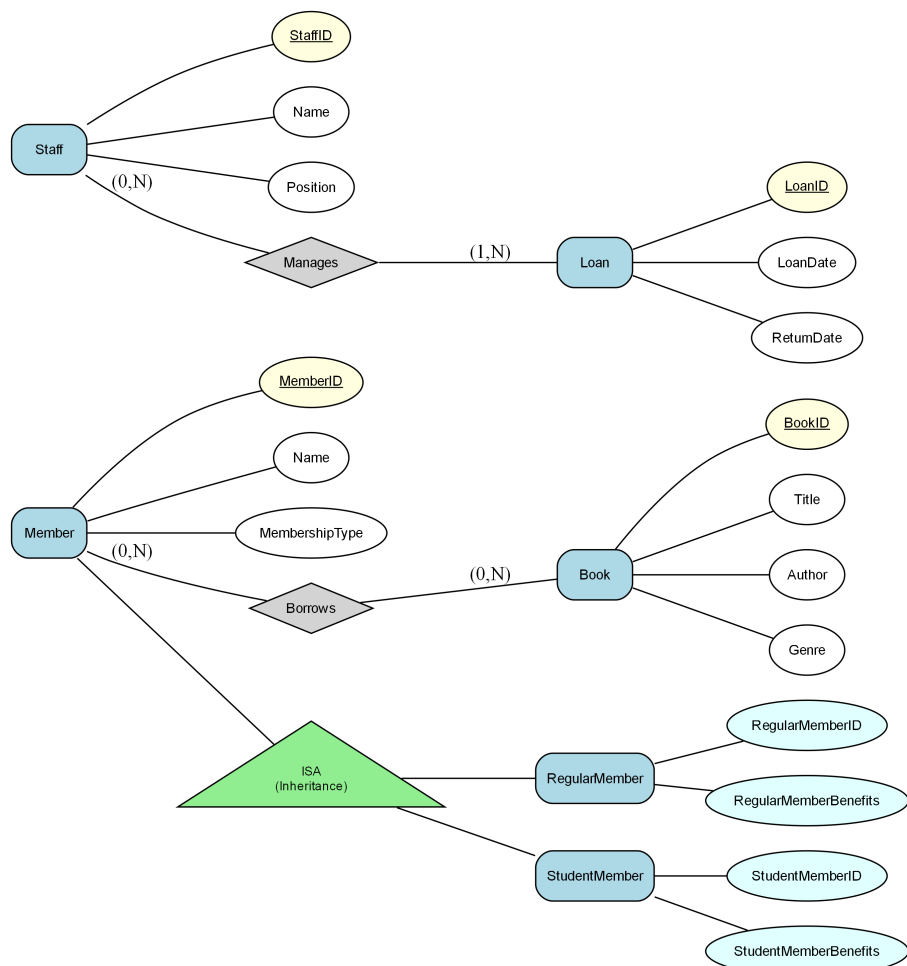
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A library management system involves several entities including: Book (attributes: BookID, Title, Author, Genre), Member (attributes: MemberID, Name, MembershipType), Loan (attributes: LoanID, LoanDate, ReturnDate), and Staff (attributes: StaffID, Name, Position). The relationships include: a Member can borrow multiple Books, a Book can be loaned out multiple times, and a Staff member manages Loans. Moreover, the MembershipType indicates whether a member is a Regular or Student member, hence creating an ISA hierarchy between Member and its subtypes. Each Loan record is associated with a LoanDate and ReturnDate

Figure: EER



- Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R : $F = \{ \{AB \rightarrow C, BC \rightarrow D, A \rightarrow E, D \rightarrow B\} \}$. Analyze the relation to identify the candidate keys, and determine its normal form.

- Find all the keys in relation R using the attribute closure method. (4 marks)
- Is R in 3NF? Give reasons for your conclusion. (3 marks)
- Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A university is designing a Database Management System to maintain student records, course enrollment, and grades. The main relation will be 'Students(sid: string, name: string, major: string, age: integer, gpa: real)'. Additionally, they will manage course details in 'Courses(cid: string, title: string, credits: integer, instructor: string)'. The objective is to ensure data integrity while supporting frequent

queries and updates.

a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

b) Code Segment:

```
```sql
SELECT * FROM Courses
WHERE PatientID = 'P001';
```
```

Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

c) Briefly explain the options available for defining and enforcing integrity constraints within the Students relation to ensure data accuracy. (2 marks)

d) Briefly explain how transactions are managed to ensure data consistency while updating records in both the Students and Courses relations. (3 marks)

e) The university decides to implement a new feature that allows students to enroll in courses but prevents them from exceeding a certain credit limit per semester. Write a T-SQL statement to set this restriction in the Courses relation.

- i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)
- ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
- iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the transactions database using a user-defined server role. (5 marks)
- iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)
- v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the client accounts and transactions databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a Library: Book (bookId: int, title: varchar(100), author: varchar(100), isbn: varchar(20), publicationYear: int, genre: varchar(50), availableCopies: int) Member (memberId: int, firstName: varchar(50), lastName: varchar(50), email: varchar(50), phone: int, address: varchar(100)) Loan (loanId: int, bookId: int, memberId: int, loanDate: date, dueDate: date, returnDate: date) Fine (fineId: int, memberId: int, amount: real, paymentStatus: varchar(50)) The 'Book' table stores information about books available in the library, including unique book ID, title, author, ISBN, publication year, genre, and the number of available copies. The 'Member' table holds details about library members, such as their unique member ID, first name, last name, email, phone number, and address. The 'Loan' table manages book loans, with each loan having a unique ID and being associated with a specific book and member. It records the loan date, due date, and return date. The 'Fine' table contains information about fines incurred by members for late returns or other penalties. It includes a fine ID, member ID, fine amount, and payment status.

a) Write SQL Queries to perform the following:

- i. Find the title of books that match specific conditions. (4 marks)
- ii. Find books that match specific criteria using WHERE clauses. (4 marks)
- iii. Find the titles and related information from books, varchars, and varchars using joins. (4 marks)

b) Create a SQL query with appropriate joins between multiple tables. (13 marks)

c) Create a trigger that automatically updates the 'Fine' column in the 'Member' table whenever a new fine is added, or an existing fine is updated. The 'Fine' column should contain the total fine amount for each member. Ensure the trigger adjusts the 'Fine' column accurately to reflect any partial payments made by members. The member pays only 50% of the total fine amount as an initial payment when the payment status is marked as 'Partial'. (15 marks)